As we've seen, a spatially flat universe requires the total mass/energy density to equal the critical density. With ordinary and dark matter contributing about 30 percent and dark energy contributing about 70 percent, everything hangs together impressively.

A more direct confirmation of the supernova results is the goal of the SuperNova/Acceleration Probe (SNAP). Proposed by scientists at the Lawrence Berkeley Laboratory, SNAP would be a satellite-borne orbiting telescope with the capacity to observe and measure more than twenty times the number of supernovae studied so far. Not only would SNAP be able to confirm the earlier result that 70 percent of the universe is dark energy, but it should also be able to determine the nature of the dark energy more precisely.

You see, although I have described the dark energy as being a version of Einstein's cosmological constant—a constant, unchanging energy that pushes space to expand—there is a closely related but alternative possibility. Remember from our discussion of inflationary cosmology (and the jumping frog) that a field whose value is perched above its lowest energy configuration can act like a cosmological constant, driving an accelerated expansion of space, but will typically do so only for a short time. Sooner or later, the field will find its way to the bottom of its potential energy bowl, and the outward push will disappear. In inflationary cosmology, this happens in a tiny fraction of a second. But by introducing a new field and by carefully choosing its potential energy shape, physicists have found ways for the accelerated expansion to be far milder in its outward push but to last far longer—for the field to drive a comparatively slow and steady accelerated phase of spatial expansion that lasts not for a fraction of a second, but for billions of years, as the field slowly rolls to the lowest energy value. This raises the possibility that, right now, we may be experiencing an extremely gentle version of the inflationary burst believed to have happened during the universe's earliest moments.

The difference between a true cosmological constant and the latter possibility, known as *quintessence*, is of minimal importance today, but has a profound effect on the long-term future of the universe. A cosmological constant is *constant*—it provides a never-ending accelerated expansion, so the universe will expand ever more quickly